Black holes—page 1

Name:

Skills: interpreting, applying, analysing, evaluating

Black holes are incredibly dense, far denser than the closely packed neutrons that make up neutron stars. Whatever form the matter inside a black hole takes, there cannot be atoms as we understand them. The mass of a black hole is thus very large for its size. For example, a black hole with the same mass as the Earth would be less than two centimetres across! All that mass creates a fiercely strong gravitational field in the space around a black hole.

The strength of any gravitational field decreases in what is known as an 'inverse square' relationship. Thus, if you double your distance from the centre of a black hole, the gravitational field strength decreases to a quarter. This means that the gravitational field near a small black hole weakens very rapidly with distance. This is different from our experience in Earth's gravitational field. For us, a change of several metres, or even several kilometres, does not create a noticeable change in the gravitational field. That is because our distance from the centre of the Earth is over 6000 kilometres.

Only the mass of a body, and the distance from its centre, affect the strength of its gravitational field. The radius of the Earth is about 6370 km, and its mass is about 6×10^{21} tonnes. The table below shows how the gravitational field strength changes with distance near the Earth's surface, and near a black hole having the same mass as the Earth.

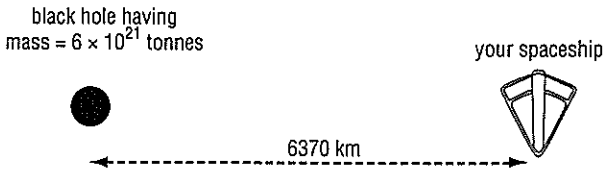
Earth's gravitational field strength, in m s^{-2}	At the Earth's surface g = 9.8299	At a height of 2 m above the surface g = 9.8299	At a height of 1 km above the surface g = 9.8268	At a height of 100 km above the surface g = 9.5284	At a height of 1000 km above the surface g = 7.0286
Gravitational field strength near a black hole, in m s^{-2}	At the black hole's surface g = 4.924×10^{18}	At a height of 2 m above the surface g = 9.883×10^{13}	At a height of 1 km above the surface g = 3.987×10^{8}	At a height of 100 km above the surface g = 39 866	At a height of 1000 km above the surface g = 398.66

If you were near a small black hole—say, a centimetre or two across—the difference between the gravitational fields at your feet and at your head would be large enough to literally pull you apart—a phenomenon called 'spaghettification'.

Questions

1 Imagine you were in a spaceship, one Earth radius away from a small black hole having the same mass as the Earth.

a Circle the correct answer:



Would you expect the gravitational field strength at the spaceship's location to be:

a more than 9.8 m s^{-2}?
b about 9.8 m s^{-2}?
c less than 9.8 m s^{-2}?
d zero?

Black holes—page 2

Name:

b Explain your answer to a.

c Would you be in danger of spaghettification at this location? Explain.

2 In some science fiction movies and stories, black holes are used as 'portals' through which travellers can reach other times and places quickly. Could this be possible? Explain.

3 Most, perhaps all, galaxies seem to have a supermassive black hole at their centre. Our own Milky Way galaxy apparently has a central black hole with a mass equivalent to millions of stars. Such black holes are detectable by strong, intermittent X-ray emissions from matter being drawn into them. X-rays are produced when very high-speed particles collide. Suggest why the X-ray emissions from around black holes are intermittent.

4 A black hole is 'black' because its escape velocity—the speed that an object needs to achieve so that it does not fall back into the black hole—exceeds the speed of light. Hence, not even light travels fast enough to get away. Despite this, individual particles may get away, a bit like molecules leaving a liquid by evaporation. Very small black holes are particularly liable to disappear by this process. Why might larger black holes not have the same risk of 'evaporating'?